INCH-POUND

MIL-C-44349(GL)
9 December 1988

MILITARY SPECIFICATION

CUP, WATER CANTEEN, COLD WEATHER, CRS

This specification is approved for use by the U.S. Army Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type and size of corrosion-resisting steel canteen cup with a folding wire handle having an 18 ounce capacity.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-P-35	-	Passivation Treatments for Corrosion-Resisting Steel
PPP-B-26	-	Bag, Plastic, (General Purpose)
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8465

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MILITARY

MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-10-450 - Cup, Canteen; Assembly (Canteen, Cold Weather, CRS)
2-10-451 - Cup, Canteen
2-10-452 - Handle, Left Hand
2-10-453 - Handle, Right Hand
2-10-454 - Hinge, Handle

(Copies of drawings required by contractors are available from the U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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of the entity controlling the document. A Non-Government (FSSM)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging
E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Nongovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Guide samples. Guide samples, when furnished, are solely for guidance and information (see 6.4). Variations from this specification may appear in the sample, in which case this specification shall govern.

3.3 Materials and components. The materials shall be as specified herein and on Drawings 2-10-451, 2-10-452, 2-10-453, and 2-10-454. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this specification.

3.3.1 Recycled materials. Recycled materials may be used in the fabrication of the cup provided that they meet the requirements of this specification.

3.4 Design and construction. The design and construction of the cup shall conform to Drawing 2-10-450 and the requirements specified herein.

3.4.1 Body. The body of the cup shall be fabricated of the corrosion-resisting steel specified on Drawing 2-10-451 and shall be of seamless construction. The top edge shall be free of sharp edges, cracks, burrs, slivers, or rough surfaces. After forming, the vertical walls of the body shall have a minimum hardness of 46 on the 30N Rockwell scale when tested as specified in 4.5.5.

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3.4.2 Handles. The handles shall be fabricated of corrosion-resisting steel and welded in accordance with Drawings 2-10-452 and 2-10-453. The welds of the handle shall withstand a minimum total force of 235 pounds when tested as specified in 4.5.4 The handle shall conform to the design, dimension, and detail requirements of Drawings 2-10-452 and 2-10-453. The handle shall fit snugly against the cup within 1/8 inch when folded and the right handle shall nest and lock into the left handle when in use.

3.4.3 Hinge. The hinge shall be fabricated of corrosion-resisting steel and shall conform to the design, dimension, and details specified on Drawing 2-10-454. The hinge shall be attached by spot welds or projection welds, shall be located on the cup, and shall be attached as shown on Drawing 2-10-450. The welds shall be free of splatter, burn holes, cracks or fractures, and discoloration and shall be tightly joined when tested as specified in 4.5.1, 4.5.2, or 4.5.3, as applicable. The hinge shall be a close fit to the contour of the body and there shall be no gaps between edge of hinge and body greater than 1/64 inch.

* 3.5 Finish. The cup shall be given a uniform frosted lusterless finish approximating a commercial 2D sheet finish and passivated in accordance with QQ-P-35. There shall be no evidence of rust, staining, pitting, or spotting when tested as specified in 4.5.6.

3.6 Marking. The letters US, the year of manufacture, and the manufacturer's name or trademark shall be stamped on the hinge as shown on Drawing 2-10-451.

3.7 Workmanship. The cup shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels. The finished cup shall be clean and free of any dent or being bent out of shape, stain, ripple, fracture, or die or draw mark. The handles and hinge shall also be free of any sharp edge, burr, or sliver. Indentation on the inside of cup due to welding of the hinge is acceptable providing the indentation is smooth and free from sharp edges, weld splatter, burr, crack, or burn through area, and does not exceed 0.005 inch in depth.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested as specified in 4.5.2 through 4.5.6, as applicable.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection.

4.4.2.1 In-process examination. In-process examination shall be made to establish conformance to the welding on wire handles requirements of Drawing 2-10-452 and 2-10-453. Whenever nonconformance is noted, correction shall be made to the applicable process and to all items processed. Items which cannot be corrected shall be removed from production.

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4.4.2.2 In-process spot or projection weld testing. The spot or projection weld sample peel test shall be conducted on two test specimens per welding machine rather than sample cups. The spot or projection welding testing shall be performed as specified in 4.5.1 at the start of each day's shifts production. Failure of test shall require correction of the welding process.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of cups. The sample unit shall be one cup. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Not a uniform, frosted, lusterless finish approximating a commercial 2D sheet finish		201
	Evidence of rust, staining, pitting, or spotting (slight water spotting, easily removed by washing shall not be considered a defect)	101	
Construction and workmanship (applicable to all components and assemblies)	Seam, lap, lamination or crease, or metal ripple	102	
	Any component missing	103	
	Component dented		202
	Component fractured	104	
	Component bent out of shape		203
	Rim of cup contains sharp edge, burr, crack, sliver, or rough spot	105	
	Evidence of die or draw mark		204
	Handle does not fold around cup within 1/8 inch as specified, or right handle does not nest and lock into left handle	106	
	Hinge plate is not close fit against body, i.e. contains gap along edge greater than 1/64 inch (use feeler gage)		205
	Handle binds or requires force in opening or folding		206
	Handle or hinge contains sharp edge, burr, or sliver	107	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Welds	Contains crack or fracture	108	
	Not number specified	109	
	Not in location shown on drawing		207
	Contains burn hole, splatter, or discoloration	110	
	Evidence of poor fusion	111	
	Any missing weld	112	
	Weld on handle not neatly performed, i.e. contains rough or sharp projection or bead		208
	Indentation in weld area on inside of cup (indentation on inside of the cup due to welding of the hinge is acceptable providing the indentation is smooth and free from sharp edges, weld splatter, burr, crack, burn through area, and does not exceed 0.005 inch in depth)		209
Marking	Missing, illegible, or incomplete		210

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on drawings that can be determined on the end items without damaging or disassembling the end items. The lot size shall be expressed in units of cups. The sample unit shall be one cup. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5 End item testing. The end items shall be tested as specified in 4.5.2 or 4.5.3, 4.5.4, 4.5.5, and 4.5.6. The lot size shall be expressed in units of cups. The sample unit shall be one cup. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container Pads not placed between layers
Content	Number per container is more or less than required Cups not arranged in container as specified

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Spot or projection weld sample peel test. Each spot or projection weld test specimen shall be of the type and thickness of material used in the cup body and hinge. Spot or projection weld each specimen as shown on Drawing 2-10-450 and then place in a vise or jig. Bend the parent metal piece and peel the hinge piece with pliers or other suitable tool. Inability of any spot weld to pull a 5/32 inch (minimum) hole in either piece of material shall constitute failure of this test. Failures of the projection welds or lack of fusion under the projection weld shall constitute failure of the test. It is not necessary to pull buttons in this test, provided that at least 3/4 of the area of each individual projection weld shows positive fusion.

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4.5.2 Spot weld cup peel test. Place the cup in a jig or vise and peel the hinge from the cup. Inability of each weld to leave a 5/32 inch (minimum) hole in either the cup or hinge shall constitute a defect.

4.5.3 Projection weld cup peel test. Separate the weldment of the cup and hinge by driving a chisel between welded components in an unwelded area or by peeling the hinge back against the weld until failure occurs. Failure outside the periphery of the weld area shall be considered evidence that the welds are satisfactory. Lack of fusion under the projection weld shall constitute a defect. It is not necessary to pull buttons in this test, provided that at least 3/4 of the area of each individual projection weld shows positive fusion.

4.5.4 Test for welds of the handle. Place the cup in a jig or vise and exert a force against both welds of the handle. Inability of any weld to withstand the minimum force specified in 3.4.2 shall constitute a defect.

4.5.5 Rockwell hardness test. Test the Rockwell hardness of the vertical walls of the body of the cup in accordance with ASTM E 18. Take four Rockwell hardness readings on the vertical walls each 90 degrees apart around the cup and approximately half way between the top and the bottom of the cup. Any sample cup having one or more readings less than 46 on the Rockwell 30N scale as specified in 3.4.1 shall constitute a defect.

4.5.6 Finish test. Expose the complete assembled cup to 95 to 100 percent relative humidity at $72^{o} \pm 2^{o}F$ for not less than 16 hours to determine compliance with rust, staining, pitting, and spotting requirements of 3.5. Inability of the cup to conform to requirements of 3.5 shall constitute a defect.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Five cups, with handles closed, shall be unit packed in a plastic bag conforming to type I or II of PPP-B-26. The bag shall be closed in accordance with style 2 or 3 of PPP-B-26.

5.1.2 Commercial. The cups shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Sixty cups, preserved as specified in 5.1, shall be packed in a shipping container. The shipping container shall be a fiberboard box conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each box shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. The cups shall be packed four in length,

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five in width, and three in depth within the box. A fiberboard pad shall be placed between each layer and on the top layer. Inside dimensions of the box shall be approximately 25 inches in length, 13-3/4 inches in width, and 10 inches in depth. Each box shall be closed, waterproofed, and reinforced with strapping or tape banding in accordance with the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6. Boxes shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Sixty cups, preserved as specified in 5.1, shall be packed in a shipping container. The shipping container shall be a fiberboard box conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each box shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. The cups shall be packed four in length, five in width, and three in depth within the box. A fiberboard pad shall be placed between each layer and on the top layer. Inside dimensions of the box shall be approximately 25 inches in length, 13-3/4 inches in width, and 10 inches in depth. Each box shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.6.

5.2.3 Commercial. Cups, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), cups packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV , or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 91 in accordance with the appendix of MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The corrosion-resisting steel cup is for use with the Carrier, Water Canteen, Cold Weather, CRS, MIL-C-44347(GL) and Canteen, Water, Cold Weather, CRS MIL-C-44348(GL). The cup is used for holding hot and cold liquids, soups, and stews in the field.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Levels of preservation and packing (see 5.1 and 5.2).
e. Type and class of unit load (see 5.2.1).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
g. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples, address the contracting office issuing the invitation for bids.

6.5 Hydroforming. Hydroforming has been found to produce acceptable cups.

6.6 Subject term (key word) listing.

Corrosion-resisting steel

Custodian:
Army - GL

Preparing activity:
Army - GL

(Project 8465-A007)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-44349(GL)	Cup, Water Canteen, Cold Weather, CRS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP-1, Feb 86

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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